

TOPIC -- CROSS ENTROPY LOSS

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$$J(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N H(p_i, q_i) = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)]$$

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Motivation In information theory, the Kraft–McMillan theorem establishes that any directly decodable coding scheme for coding a message to identify one value x_i out of a set of possibilities $\{x_1, \dots, x_n\}$ can be seen as representing an implicit probability distribution $q(x_i) = \frac{1}{2^{\ell_i}}$ over $\{x_1, \dots, x_n\}$, where ℓ_i is the length of the code for x_i in bits. Therefore, cross-entropy can be interpreted as the expected message-length per datum when a wrong distribution q is assumed while the data actually follows a distribution p . That is why the expectation is taken over the true probability distribution p and not q . Indeed the expected message-length under the true distribution p is

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$$\frac{\partial}{\partial w_{ij}} \ln \hat{y}^i = \frac{\partial}{\partial w_{ij}} \ln \frac{1}{1 + e^{-w_{ij}x_{ij} + k_j}} = x_{ij} e^{-w_{ij}x_{ij} + k_j} \frac{1}{1 + e^{-w_{ij}x_{ij} + k_j}} = x_{ij} (1 - \hat{y}^i),$$

$$\frac{\partial}{\partial w_{ij}} \ln \frac{1}{1 + e^{-w_{ij}x_{ij} + k_j}} = \frac{\partial}{\partial w_{ij}} \ln \frac{1}{1 + e^{-w_{ij}x_{ij} + k_j}} = x_{ij} (1 - \hat{y}^i).$$

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The situation for continuous distributions is analogous. We have to assume that p and q are absolutely continuous with respect to some reference measure r (usually r is a Lebesgue measure on a Borel σ -algebra). Let P and Q be probability density functions of p and q with respect to r . Then

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$$H(p, q) = - \sum_m p_m \log q_m = -y \log \hat{y} - (1-y) \log (1 - \hat{y}).$$

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$$e^k = H(p, q^k) - \lambda \sum_{j \neq k} H(q^j, q^k)$$

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Writing our likelihood function as the product of observations from the distribution q_{θ} :